

	1	2	3	4	5	6	7	8					
A	DEEPREAL MAIN PCBA – ELECTRICAL CAPTURE MAP 15 pages total. This hierarchy defines what must be captured; it is not an ERC-clean production schematic. Red gates: i.MX95 / LPDDR4X / eMMC vendor pin maps, Lattice compile, flex orientation and power calculations. USB–C_PD_and_VBUS File: 02_usb_c_pd_and_vbus.kicad_sch System_and_motor_bucks File: 03_system_and_motor_bucks.kicad_sch PF09_Pf53_power_tree File: 04_pf09_pf53_power_tree.kicad_sch												
B	iMX95_power_and_decoupling File: 05_imx95_power_and_decoupling.kicad_sch	iMX95_boot_clock_debug File: 06_imx95_boot_clock_debug.kicad_sch	LPDDR4X File: 07_lpddr4x.kicad_sch										
C	eMMC_and_board_identity File: 08_emmc_and_board_identity.kicad_sch	Face_camera_bridge File: 09_face_camera_bridge.kicad_sch	Interaction_camera_bridge File: 10_interaction_camera_bridge.kicad_sch										
	Face_head_interface File: 11_face_head_interface.kicad_sch	Interaction_head_interface File: 12_interaction_head_interface.kicad_sch	Motors_and_absolute_angle File: 13_motors_and_absolute_angle.kicad_sch										
D	Audio_tamper_thermal_misc File: 14_audio_tamper_thermal_misc.kicad_sch	Test_and_manufacturing File: 15_test_and_manufacturing.kicad_sch											
E													
F	1	2	3	4	5	6	7	8					

PRELIMINARY / NOT FOR FABRICATION
90 x 28 mm / 10-layer HDI working assumption
DeepReal
Sheet: /
File: deepreal-main-pcba.kicad_sch

Title: DeepReal Main PCBA – Cover and Constraints

Size: A3

Date:

Rev: 0.2–capture

KiCad E.D.A. 10.0.6

Id: 1/15

PRELIMINARY / NOT FOR FABRICATION 90 x 28 mm / 10–layer HDI working assumption		
DeepReal		
Sheet: / File: deepreal–main–pcba.kicad_sch		
Title: DeepReal Main PCBA – Cover and Constraints		
Size: A3	Date:	Rev: 0.2–capture
KiCad E.D.A. 10.0.6	Id: 1/15	

	1	2	3	4	5	6						
A	USB-C PD and VBUS					A						
	STATUS: DEFINED											
	PRIMARY REFERENCES: J1 U10 U11 D1 D2 D9											
	CLOSURE EVIDENCE: Official symbols; CC configuration; protection review											
	ARCHITECTURE DEFINED – component values and pin-level review still required											
B	See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B						
C						C						
D	PRELIMINARY / NOT FOR FABRICATION Page 2/15 – DEFINED DeepReal Sheet: /USB-C_PD_and_VBUS/ File: 02_usb_c_pd_and_vbus.kicad_sch Title: USB-C PD and VBUS <table><tr><td>Size: A4</td><td>Date:</td><td>Rev: 0.2-capture</td></tr><tr><td>KiCad E.D.A. 10.0.6</td><td></td><td>Id: 2/15</td></tr></table>					Size: A4	Date:	Rev: 0.2-capture	KiCad E.D.A. 10.0.6		Id: 2/15	D
Size: A4	Date:	Rev: 0.2-capture										
KiCad E.D.A. 10.0.6		Id: 2/15										
	1	2	3	4	5	6						

	1	2	3	4	5	6						
A	System and motor bucks STATUS: DEFINED PRIMARY REFERENCES: U8 U9 L1 L2 CLOSURE EVIDENCE: Calculated magnetics; compensation; thermal review ARCHITECTURE DEFINED – component values and pin-level review still required See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					A						
B						B						
C						C						
D	PRELIMINARY / NOT FOR FABRICATION Page 3/15 – DEFINED DeepReal Sheet: /System_and_motor_bucks/ File: 03_system_and_motor_bucks.kicad_sch Title: System and motor bucks <table><tr><td>Size: A4</td><td>Date:</td><td>Rev: 0.2-capture</td></tr><tr><td>KiCad E.D.A. 10.0.6</td><td></td><td>Id: 3/15</td></tr></table>					Size: A4	Date:	Rev: 0.2-capture	KiCad E.D.A. 10.0.6		Id: 3/15	D
Size: A4	Date:	Rev: 0.2-capture										
KiCad E.D.A. 10.0.6		Id: 3/15										
	1	2	3	4	5	6						

1	2	3	4	5	6							
A	PF09 PF53 power tree STATUS: DEFINED PRIMARY REFERENCES: U5 U6 U7 L3–L9 CLOSURE EVIDENCE: NXP OTP/sequence review; rail calculation ARCHITECTURE DEFINED – component values and pin–level review still required See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					A						
B						B						
C						C						
D	PRELIMINARY / NOT FOR FABRICATION Page 4/15 – DEFINED DeepReal Sheet: /PF09_PF53_power_tree/ File: 04_pf09_pf53_power_tree.kicad_sch Title: PF09 PF53 power tree<table><tr><td>Size: A4</td><td>Date:</td><td>Rev: 0.2–capture</td></tr><tr><td>KiCad E.D.A. 10.0.6</td><td></td><td>Id: 4/15</td></tr></table>					Size: A4	Date:	Rev: 0.2–capture	KiCad E.D.A. 10.0.6		Id: 4/15	D
Size: A4	Date:	Rev: 0.2–capture										
KiCad E.D.A. 10.0.6		Id: 4/15										
1	2	3	4	5	6							

	1	2	3	4	5	6						
A	iMX95 power and decoupling STATUS: PINMAP_GATE PRIMARY REFERENCES: U1 C120–C219 CLOSURE EVIDENCE: Official 548–ball symbol; power audit PIN–LEVEL CAPTURE BLOCKED – import and audit official package data See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					A						
B						B						
C						C						
D	PRELIMINARY / NOT FOR FABRICATION Page 5/15 – PINMAP_GATE DeepReal Sheet: /iMX95_power_and_decoupling/ File: 05_imx95_power_and_decoupling.kicad_sch Title: iMX95 power and decoupling <table><tr><td>Size: A4</td><td>Date:</td><td>Rev: 0.2–capture</td></tr><tr><td>KiCad E.D.A. 10.0.6</td><td></td><td>Id: 5/15</td></tr></table>					Size: A4	Date:	Rev: 0.2–capture	KiCad E.D.A. 10.0.6		Id: 5/15	D
Size: A4	Date:	Rev: 0.2–capture										
KiCad E.D.A. 10.0.6		Id: 5/15										
	1	2	3	4	5	6						

	1	2	3	4	5	6
A	iMX95 boot clock debug					A
	STATUS: PINMAP_GATE					
	PRIMARY REFERENCES: U1 Y1 J6					
	CLOSURE EVIDENCE: Boot table; reset timing; debug review					
	PIN-LEVEL CAPTURE BLOCKED – import and audit official package data					
B	See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 6/15 – PINMAP_GATE DeepReal Sheet: /iMX95_boot_clock_debug/ File: 06_imx95_boot_clock_debug.kicad_sch Title: iMX95 boot clock debug Size: A4 Date: Rev: 0.2-capture KiCad E.D.A. 10.0.6 Id: 6/15					D
	1	2	3	4	5	6

	1	2	3	4	5	6						
A	LPDDR4X					A						
	STATUS: DDR_GATE											
	PRIMARY REFERENCES: U1 U4 C220–C259											
	CLOSURE EVIDENCE: NXP DDR tool; Micron pin map; topology review											
	PIN–LEVEL CAPTURE BLOCKED – import and audit official package data											
B	See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B						
C						C						
D	PRELIMINARY / NOT FOR FABRICATION Page 7/15 – DDR_GATE DeepReal Sheet: /LPDDR4X/ File: 07_lpddr4x.kicad_sch Title: LPDDR4X <table><tr><td>Size: A4</td><td>Date:</td><td>Rev: 0.2–capture</td></tr><tr><td>KiCad E.D.A. 10.0.6</td><td></td><td>Id: 7/15</td></tr></table>					Size: A4	Date:	Rev: 0.2–capture	KiCad E.D.A. 10.0.6		Id: 7/15	D
Size: A4	Date:	Rev: 0.2–capture										
KiCad E.D.A. 10.0.6		Id: 7/15										
	1	2	3	4	5	6						

	1	2	3	4	5	6
A	eMMC and board identity					A
	STATUS: PINMAP_GATE					
	PRIMARY REFERENCES: U15 U16					
	CLOSURE EVIDENCE: Micron pin map; boot test; EEPROM address audit					
	PIN-LEVEL CAPTURE BLOCKED – import and audit official package data					
B	See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 8/15 – PINMAP_GATE DeepReal Sheet: /eMMC_and_board_identity/ File: 08_emmc_and_board_identity.kicad_sch Title: eMMC and board identity Size: A4 Date: Rev: 0.2-capture KiCad E.D.A. 10.0.6 Id: 8/15					D
	1	2	3	4	5	6

	1	2	3	4	5	6
A	Face camera bridge					A
	STATUS: FPGA_GATE					
	PRIMARY REFERENCES: U2 D3~D5					
	CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment					
	PIN-LEVEL CAPTURE BLOCKED – import and audit official package data					
B	See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 9/15 – FPGA_GATE DeepReal Sheet: /Face_camera_bridge/ File: 09_face_camera_bridge.kicad_sch Title: Face camera bridge Size: A4 Date: Rev: 0.2-capture KiCad E.D.A. 10.0.6 Id: 9/15					D
	1	2	3	4	5	6

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

12345

A

Interaction camera bridge

STATUS: FPGA_GATE

PRIMARY REFERENCES: U3 D6–D8

CLOSURE EVIDENCE: Radiant compile; csfBGA121 legal pin assignment

PIN–LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 10/15 – FPGA_GATE
DeepReal

Sheet: /Interaction_camera_bridge/
File: 10_interaction_camera_bridge.kicad_sch

Title: Interaction camera bridge

Size: A4Date:Rev: 0.2–capture

KiCad E.D.A. 10.0.6Id: 10/15

12345

D

Id: 10/15

	1	2	3	4	5	6
A	Face head interface					A
	STATUS: FLEX_GATE					
	PRIMARY REFERENCES: J2 U21					
	CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review					
	PIN-LEVEL CAPTURE BLOCKED – import and audit official package data					
B	See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 11/15 – FLEX_GATE DeepReal Sheet: /Face_head_interface/ File: 11_face_head_interface.kicad_sch Title: Face head interface Size: A4 Date: Rev: 0.2-capture KiCad E.D.A. 10.0.6 Id: 11/15					D
	1	2	3	4	5	6

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

12

Interaction head interface

STATUS: FLEX_GATE

PRIMARY REFERENCES: J3 U22

CLOSURE EVIDENCE: 51-pin flex drawing; impedance and current review

PIN-LEVEL CAPTURE BLOCKED – import and audit official package data

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 12/15 – FLEX_GATE
DeepReal

Sheet: /Interaction_head_interface/
File: 12_interaction_head_interface.kicad_sch

Title: Interaction head interface

Size: A4Date:Rev: 0.2-capture

KiCad E.D.A. 10.0.6Id: 12/15

Id: 12/15

	1	2	3	4	5	6
A	Motors and absolute angle STATUS: DEFINED PRIMARY REFERENCES: U13 U14 U18 U19 J4 J5 CLOSURE EVIDENCE: Current limit; stall; encoder and ESD review ARCHITECTURE DEFINED – component values and pin-level review still required See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					A
B						B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 13/15 – DEFINED DeepReal Sheet: /Motors_and_absolute_angle/ File: 13_motors_and_absolute_angle.kicad_sch Title: Motors and absolute angle Size: A4 Date: Rev: 0.2-capture KiCad E.D.A. 10.0.6 Id: 13/15					D
	1	2	3	4	5	6

Audio_tamper_thermal_misc

STATUS: DEFINED

PRIMARY REFERENCES: MK1 SW1 U17 U20

CLOSURE EVIDENCE: Acoustic keep-out; tamper fail state; address audit

ARCHITECTURE DEFINED – component values and pin-level review still required

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 14/15 – DEFINED
DeepReal
Sheet: /Audio_tamper_thermal_misc/
File: 14_audio_tamper_thermal_misc.kicad_sch
Title: Audio_tamper_thermal_misc
Size: A4 Date: Rev: 0.2-capture
KiCad E.D.A. 10.0.6 Id: 14/15

Audio tamper thermal misc

STATUS: DEFINED

PRIMARY REFERENCES: MK1 SW1 U17 U20

CLOSURE EVIDENCE: Acoustic keep-out; tamper fail state; address audit

ARCHITECTURE DEFINED – component values and pin-level review still required

See circuit-specification.md, net-registry.csv, component-register.csv and pinmap-gates.csv.
Do not add copper or release fabrication data until this page is electrically captured and reviewed.

PRELIMINARY / NOT FOR FABRICATION
Page 14/15 – DEFINED
DeepReal
Sheet: /Audio_tamper_thermal_misc/
File: 14_audio_tamper_thermal_misc.kicad_sch
Title: Audio tamper thermal misc
Size: A4 Date: Rev: 0.2–capture
KiCad E.D.A. 10.0.6 Id: 14/15

Id: 14/15

	1	2	3	4	5	6
A	Test and manufacturing					A
	STATUS: DEFINED					
	PRIMARY REFERENCES: J6 TP1–TP32					
	CLOSURE EVIDENCE: DFT review; no–fit list; rail–access review					
	ARCHITECTURE DEFINED – component values and pin–level review still required					
B	See circuit–specification.md, net–registry.csv, component–register.csv and pinmap–gates.csv. Do not add copper or release fabrication data until this page is electrically captured and reviewed.					B
C						C
D	PRELIMINARY / NOT FOR FABRICATION Page 15/15 – DEFINED DeepReal Sheet: /Test_and_manufacturing/ File: 15_test_and_manufacturing.kicad_sch Title: Test and manufacturing Size: A4 Date: Rev: 0.2–capture KiCad E.D.A. 10.0.6 Id: 15/15					D
	1	2	3	4	5	6